

SAVE, Benin

WMO: 653350

Lat: 8.033N

Lon: 2.467E

Elev: 200

StdP: 98.95

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	21.4	1.9	4.4	27.5	4.1	5.2	27.4	5.5	26.5	4.7	25.9	0.9	0	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.7	36.7	23.2	35.9	23.4	35.1	23.6	26.9	32.1	26.5	31.8	26.2	31.6	1.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	21.3	29.7	25.1	20.7	29.2	24.7	20.3	28.8	85.7	32.1	84.0	32.1	82.6	31.8	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.9	4.2	3.5	DB	18.8	38.7	1.5	0.9	17.7	39.4	16.8	39.9	16.0	40.3	14.9	41.0	(4)
(5)			WB	10.7	28.2	1.0	1.2	10.0	29.1	9.4	29.7	8.8	30.4	8.0	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	28.3	30.0	30.1	29.1	28.0	26.8	25.9	25.3	26.0	26.9	28.3	28.2	(6)	
	DBStd	1.91	1.54	1.20	1.28	1.47	1.41	1.14	1.01	0.94	0.96	1.07	1.03	1.14	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6470	567	561	623	572	557	504	493	476	481	523	548	566	(10)	
	CDD18.3	3428	308	327	364	322	299	254	235	217	231	265	298	307	(11)	
	CDH23.3	36611	3793	4399	4764	3897	3268	2352	1780	1400	1738	2375	3261	3585	(12)	
(13)	CDH26.7	16046	1923	2408	2545	1872	1359	759	414	248	437	833	1473	1774	(13)	
(14)	Wind	WSAvg	1.3	1.1	1.4	1.6	1.7	1.5	1.5	1.6	1.3	1.0	0.9	1.0	(14)	
(15) Precipitation	PrecAvg	1103	6	17	64	107	124	158	164	151	166	114	14	4	(15)	
	PrecMax	1936	68	78	181	212	243	350	670	352	342	266	132	53	(16)	
	PrecMin	640	0	0	4	11	30	51	26	28	12	9	0	0	(17)	
	PrecStd	256	13	20	43	53	51	61	112	85	68	52	25	10	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	37.9	38.1	36.7	35.1	32.7	31.1	30.8	31.6	33.1	35.2	36.0	(19)	
		MCWB	21.3	22.4	23.7	25.4	25.6	25.4	25.0	24.8	25.3	25.5	23.5	20.7	(20)	
	2%	DB	35.4	36.6	36.8	35.4	33.7	31.8	30.3	29.7	30.7	32.1	34.1	34.8	(21)	
		MCWB	21.7	22.8	23.9	25.6	25.8	25.3	24.8	24.6	25.0	25.4	23.9	21.4	(22)	
	5%	DB	34.5	35.8	35.8	34.4	32.8	31.1	29.6	28.9	29.9	31.2	33.3	33.9	(23)	
		MCWB	21.9	23.1	24.1	25.6	25.8	25.3	24.7	24.4	24.9	25.4	24.1	22.0	(24)	
	10%	DB	33.6	35.0	34.8	33.4	32.0	30.3	28.9	28.1	29.1	30.4	32.5	33.1	(25)	
		MCWB	21.8	23.4	24.5	25.6	25.8	25.1	24.4	24.3	24.8	25.3	24.2	22.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.8	26.9	27.2	27.5	26.5	26.0	25.7	26.2	27.0	27.1	25.7	(27)	
		MCDB	32.4	33.8	33.3	32.9	32.1	30.9	29.8	29.1	30.0	30.5	31.0	32.3	(28)	
	2%	WB	25.3	26.1	26.2	26.7	26.7	25.9	25.4	25.1	25.6	26.2	26.2	25.1	(29)	
		MCDB	31.3	32.8	32.9	32.6	31.7	30.5	29.2	28.5	29.5	30.3	31.0	31.4	(30)	
	5%	WB	24.8	25.6	25.9	26.3	26.3	25.6	25.0	24.7	25.2	25.8	25.7	24.7	(31)	
		MCDB	30.6	32.0	32.5	32.2	31.3	29.9	28.8	28.0	29.1	30.1	30.8	30.5	(32)	
	10%	WB	24.3	25.2	25.6	26.0	26.0	25.2	24.6	24.4	24.9	25.5	25.3	24.3	(33)	
		MCDB	29.8	31.3	31.9	31.9	30.9	29.4	28.2	27.5	28.6	29.7	30.4	29.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	10.7	10.2	9.0	7.9	6.8	5.9	5.6	6.4	7.7	9.5	11.3	(35)	
		MCDBR	12.4	11.7	11.5	10.3	9.2	8.0	7.0	7.0	7.6	8.8	10.6	12.2	(36)	
	5% WB	MCWBR	3.9	3.2	2.9	2.6	2.8	2.9	2.7	3.0	3.2	3.5	3.0	3.8	(37)	
		MCWBR	9.8	10.1	10.0	8.9	8.1	7.2	6.4	6.2	7.0	8.2	9.1	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.783	0.807	0.821	0.702	0.596	0.544	0.514	0.506	0.501	0.564	0.617	0.680	(40)	
		taud	1.534	1.515	1.486	1.708	1.903	2.048	2.159	2.175	2.191	1.980	1.815	1.697	(41)	
	Ebn at Noon		597	602	603	671	728	757	785	805	819	766	711	657	(42)	
		Edh at Noon	286	299	310	245	197	168	151	152	151	185	214	238	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.09	5.19	5.43	5.43	5.21	4.71	4.07	3.68	4.25	4.90	5.20	5.02	(44)	
	RadStd		0.14	0.18	0.21	0.18	0.19	0.24	0.19	0.24	0.24	0.18	0.14	0.17	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.28	+0.31	N/A	+0.36	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page